Windows Server 2022 Deployment & Initial Configuration

Technical Lab Report: Infrastructure Deployment

Server hostname: DC01 | **Virtual machine:** WindowsServer2022 | **Platform:** Oracle VirtualBox

Operating system: Windows Server 2022 Standard Evaluation (Desktop Experience)

Report scope: Installation, initial networking, remote administration, guest tools, and recovery snapshot

---

1. Executive Summary

Windows Server 2022 was installed as a VirtualBox guest and renamed **DC01**. A static IPv4 address was assigned to the Host-Only interface, and connectivity to the host was verified. Remote Desktop was enabled, VirtualBox Guest Additions were installed, and a baseline snapshot was created.

The operating system deployment and baseline administration tasks are complete. Active Directory Domain Services is **not yet installed**, and the Internal Network adapter is still awaiting a static address.

Objectives

- Create a dedicated Windows Server virtual machine.
- Install Windows Server 2022 with the graphical Desktop Experience.
- Set a clear server hostname for future lab administration.
- Configure the VM network interfaces and assign a static address to the Host-Only interface.
- Enable Remote Desktop for controlled administration.
- Install VirtualBox Guest Additions for improved guest integration.
- Create a snapshot that preserves a known baseline before future changes.

Scope Boundary

This deployment covers the operating system and baseline administration only. Active Directory Domain Services (AD DS), domain creation, DNS server configuration for a domain, and domain joining are **not** included in the completed work.

Documentation Note

This report intentionally contains no screenshots. Configuration details are based on the steps and command output recorded during deployment. The Internal Network interface is marked as pending because it received an automatic APIPA address during verification.

2. Environment and VM Specifications

Parameter	Recorded Configuration
Hypervisor	Oracle VirtualBox
VM name	WindowsServer2022
Guest operating system	Windows Server 2022 Standard Evaluation (Desktop Experience)
Hostname	DC01
Memory allocation	4096 MB (4 GB)
Virtual CPUs	2
Virtual disk	Approximately 50 GB, dynamically allocated VDI
Network adapters	Three: NAT, Internal Network, Host-Only
Intended future role	Active Directory lab server / potential domain controller

> **Resource note:** The host computer has 8 GB of RAM in total. Allocating 4 GB to this VM leaves limited memory for the host and other VMs, so avoid running several resource-intensive guests at the same time.

3. Installation Method

A separate virtual hard disk was created for the Windows Server VM; the existing Parrot OS disk was not reused. The Windows Server installation media was attached to the VM's virtual optical drive.

1. Created the **WindowsServer2022** VM in VirtualBox.
2. Assigned 4 GB RAM, 2 virtual CPUs, and a dynamically allocated virtual disk of approximately 50 GB.
3. Configured Adapter 1 as NAT, Adapter 2 as Internal Network (**SOC-INTERNAL**), and Adapter 3 as Host-Only.
4. Booted from the Windows Server 2022 ISO.
5. Selected the Standard Evaluation edition with Desktop Experience.
6. Chose a custom installation and installed the operating system to the new virtual disk.
7. Set the local Administrator password and signed in after installation.

The Desktop Experience edition was selected to make initial administration easier through Server Manager and the Windows networking tools.

4. Server Identity Configuration

After installation, the computer was renamed to **DC01** and restarted to apply the new name.

Setting	Value	Purpose
Computer name	DC01	Stable and recognizable server identity
Workgroup / domain	Not joined to a domain	Domain configuration is planned for a later lab
Local administration	Built-in Administrator account	Used for initial setup and configuration

> **Naming rationale:** DC01 is a conventional hostname for a first domain controller. The hostname alone does not make the server a domain controller; AD DS must be installed and configured separately.

5. Network Architecture and IP Plan

The VM uses three virtual network adapters to separate internet access, an isolated VM-to-VM network, and host-to-VM management access.

Guest Adapter	VirtualBox Mode	IPv4 / Mask	Gateway	State
Ethernet	NAT	10.0.2.15 / 24	10.0.2.2	Configured
Ethernet 2	Internal Network (SOC-INTERNAL)	169.254.13.238 / 16 (APIPA)	None	**Pending**
Ethernet 3	Host-Only Adapter	192.168.220.40 / 24 (static)	None	Configured

Adapter 1: NAT

The NAT interface received IPv4 address 10.0.2.15 with subnet mask 255.255.255.0 and default gateway 10.0.2.2. It provides outbound connectivity for tasks such as operating-system updates.

Adapter 2: Internal Network

This adapter is attached to the VirtualBox internal network named SOC-INTERNAL. During the recorded check it held address 169.254.13.238 with subnet mask 255.255.0.0. This is an Automatic Private IP Addressing (APIPA) address, normally assigned when no DHCP lease or manual IPv4 configuration is available. A static address should be configured before this interface is used for the future domain lab.

Adapter 3: Host-Only

The Host-Only interface was configured with static address 192.168.220.40 and subnet mask 255.255.255.0. The default gateway and DNS fields were left blank so that the NAT interface remains the primary route for outbound connectivity.

Connectivity Test

A ping from DC01 to the host's VirtualBox Host-Only address was successful:

```
ping 192.168.220.1
```

> **Network status:** Host-Only connectivity to 192.168.220.1 was verified. The Internal Network adapter remains unfinished and should be assigned an address in the SOC-INTERNAL subnet before domain controller configuration.

6. Remote Desktop Configuration

Remote Desktop was enabled through **Server Manager → Local Server → Remote Desktop**. The Remote Desktop service (TermService) was checked and confirmed to be running.

Verification command:

```
Get-Service TermService
```

Remote Desktop provides a graphical remote administration path and should be used only from trusted systems on the lab network. Windows Firewall should remain enabled, and inbound Remote Desktop access should be restricted to trusted networks and authorized users.

For this lab, the Host-Only network is the preferred path for connecting from the Windows host. Remote Desktop should never be exposed directly to the public internet.

7. VirtualBox Guest Additions

Guest Additions were installed after the operating system and initial configuration were complete.

1. Selected **Devices → Insert Guest Additions CD Image** in the VirtualBox VM window.
2. Opened the mounted Guest Additions CD in Windows File Explorer.
3. Ran **VBoxWindowsAdditions-amd64**.
4. Completed the installer with its default options and approved the required installation prompts.
5. Restarted Windows Server after installation.

Guest Additions provide integration features between the guest operating system and VirtualBox, such as improved mouse integration and display resizing where supported.

8. Snapshot and Recovery Point

Property	Recorded Value
Snapshot name	DC01-Base-Configured
Purpose	Baseline recovery point before future Active Directory changes
Created after	OS installation, hostname configuration, Host-Only IP setup, Remote Desktop enablement, and Guest Additions installation

The snapshot allows the VM to be returned to the captured state if later configuration changes cause problems. It is not a substitute for a separate backup, because snapshots depend on the original virtual disk and consume additional storage.

9. Validation Results

Validation Item	Result	Notes
Windows Server 2022 installed	PASS	Desktop Experience installed
Hostname set to DC01	PASS	Restarted after rename
NAT IPv4 settings	PASS	10.0.2.15; gateway 10.0.2.2
Host-Only static IPv4	PASS	192.168.220.40/24
Ping to Host-Only address	PASS	192.168.220.1 replied
Remote Desktop enabled	PASS	TermService was running
Guest Additions	PASS	Installed and VM rebooted
Baseline snapshot	PASS	DC01-Base-Configured created
Internal Network IPv4 plan	PENDING	APIPA address observed
Active Directory role	NOT IN SCOPE	Planned for a future lab

10. Outstanding Items and Next Steps

1. **Configure the Internal Network address.** Choose an unused static IP in the SOC-INTERNAL subnet after checking the addresses already assigned to Kali, Parrot, pfSense, and other lab systems.
2. **Plan DNS.** Before promoting the server to a domain controller, decide the domain name and DNS design. A domain controller typically uses a stable address and DNS configured for the domain environment.
3. **Verify network isolation and routing.** Confirm which lab interfaces should communicate and whether pfSense is expected to route between networks.
4. **Take a separate backup when appropriate.** Retain the baseline snapshot and consider a separate backup of important VM files.
5. **Continue with Active Directory in a later lab.** Install AD DS only after the internal network and DNS plan are correct.

> **Readiness assessment:** The operating system deployment and baseline management tasks are complete. The VM is not yet ready to act as a domain controller, because the Internal Network IPv4 configuration and the Active Directory / DNS setup remain outstanding.

11. Conclusion

Windows Server 2022 was deployed successfully in Oracle VirtualBox and assigned the hostname DC01. The NAT adapter was configured for outbound connectivity, the Host-Only adapter was assigned static IPv4 address 192.168.220.40/24, and a ping test to the host's Host-Only address succeeded. Remote Desktop was enabled, VirtualBox Guest Additions were installed, and the DC01-Base-Configured snapshot was created.

This provides a baseline for future Windows infrastructure practice. The next technical milestone is to configure the Internal Network adapter with an unused address in SOC-INTERNAL, validate the intended lab connectivity, and then proceed with DNS and Active Directory configuration.

Appendix A: Verification Commands

Purpose	Command
Display full network configuration	`ipconfig /all`
Verify computer name	`hostname`
Check Remote Desktop service	`Get-Service TermService`
Test Host-Only connectivity	`ping 192.168.220.1`